Mountain Glacier Change In Ecuador

ISMT E 158 Final Project Report

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Background

Sometimes referred to as the “water towers of the world”, mountain Glaciers play a key role in the Hydrological. Cycle. In the Andean region, mountains are crucial in providing water to over 75 million people within the whole region, and some 20 million more people downstream. At higher elevations, glaciers provide a steady stream of meltwater when it is most needed: during the dry season.

However, as UNESCO mentions in its Andean Glacier and Water Atlas[1], Climate Change has already been impacting the Andean Region. In particular, this report states that the annual mean temperature has increased by 0.8°C on average during the 20th century in the Tropical Andes (Venezuela, Colombia, Ecuador and Peru). Furthermore, the report points to evidence that this increase has been faster at high altitudes. And finally, gives estimations that temperatures in the region could increase even faster in the upcoming years.

Ecuador’s glaciers are restricted to the highest peaks, but they remain a vital source of water to surrounding populations. It is estimated that glacier meltwater contributes to around 5 per cent of the available water supply in Quito, the largest city. It is then important to understand the patterns and the nature of the changes in these glaciers. Local studies performed by local glaciologist reveal periods of glacier retreat as well as gains of glacier mass and positive balances, apparently closely related to temperature changes.

The highest peak in Ecuador is the Chimborazo volcano. Rising at more than 6,000 meters above sea level, it has always been considered to have a “perennial” glacier. It seemed to be the natural choice for studying the phenomenon of glacier retreat from the perspective of Remote Sensing, (besides other more “logistic” reasons that will be explained later). But it was also a choice of symbolic importance: the Chimborazo is present in the national emblem, as depicting the view that one can appreciate in cloud free days from the Coast of Ecuador. And it as also related to the former tradition of the “Ice Merchants”: people that went up onto the glacier to “harvest” ice and then sell it at the nearby towns.

Research Goals

From the previous section it seemed that the topic of glacier change could be a very interesting one to be studied from the perspective of Remote Sensing. Not only from the application of the tools and methods, but in addition, it consisted of new land cover type not directly addressed in class.

These findings about the effects of temperature increase on glaciers in the Tropical Andes made me formulate the following uqestions to be addressed using Remote Sensing:

Research Questions

1. Given that local studies report periods of glacier retreat and glacier recovery, is this retreat permanent? Or due to these cycles, can we hope for a full recovery?

2. As Climate Change is a “recent” phenomenon, attributed to the last 20 years or so, are the glaciers retreating faster than before?

Research Subject

The Chimborazo volcano was chosen, for the following reasons:

1. Glacier Relevance: largest mountain glacier in Ecuador
2. Imagery: largest set of cloud-free images encountered from LS-7 scenes where the view of the volcano is not affected by the SLC malfunction lines

Research Approach

Poner aqui el WF general.

Hablar de las herramientas usadas QGIS, EE, MODIS.

Data Set Selection

Detalle de como se buscaron y escogieron: LS, Sentinel, pb de nubes…MODIS NDSI

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RS processing & analysis

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Results

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Bibliography

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Data Sources

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